



Starlink-Based Navigation System for Fixed-Wing UAV

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System Architecture

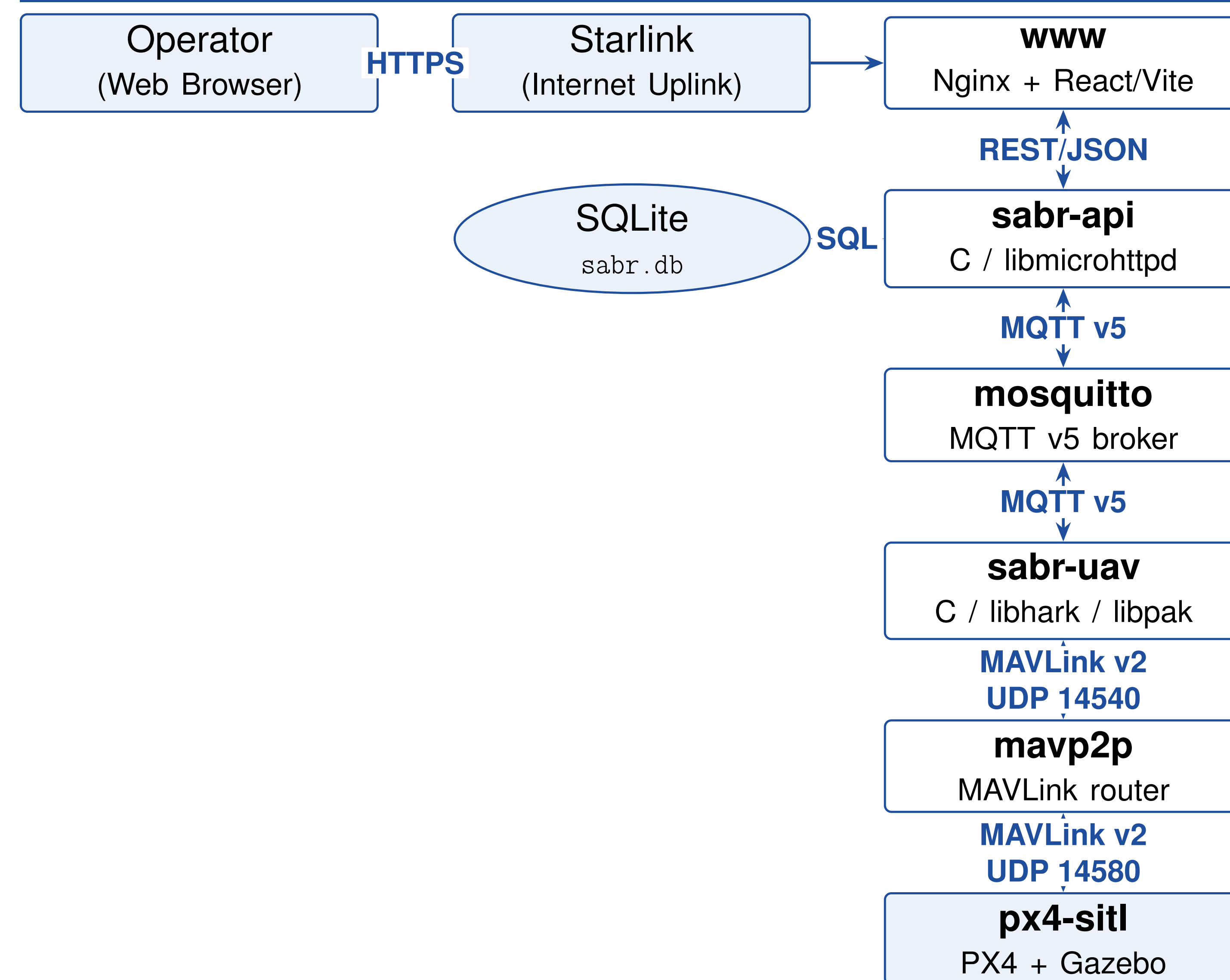


Figure 1. SABR runtime stack.
Six containerized services on the ground station. Blue boxes are externals; white boxes are SABR-authored services.

Operator Interface

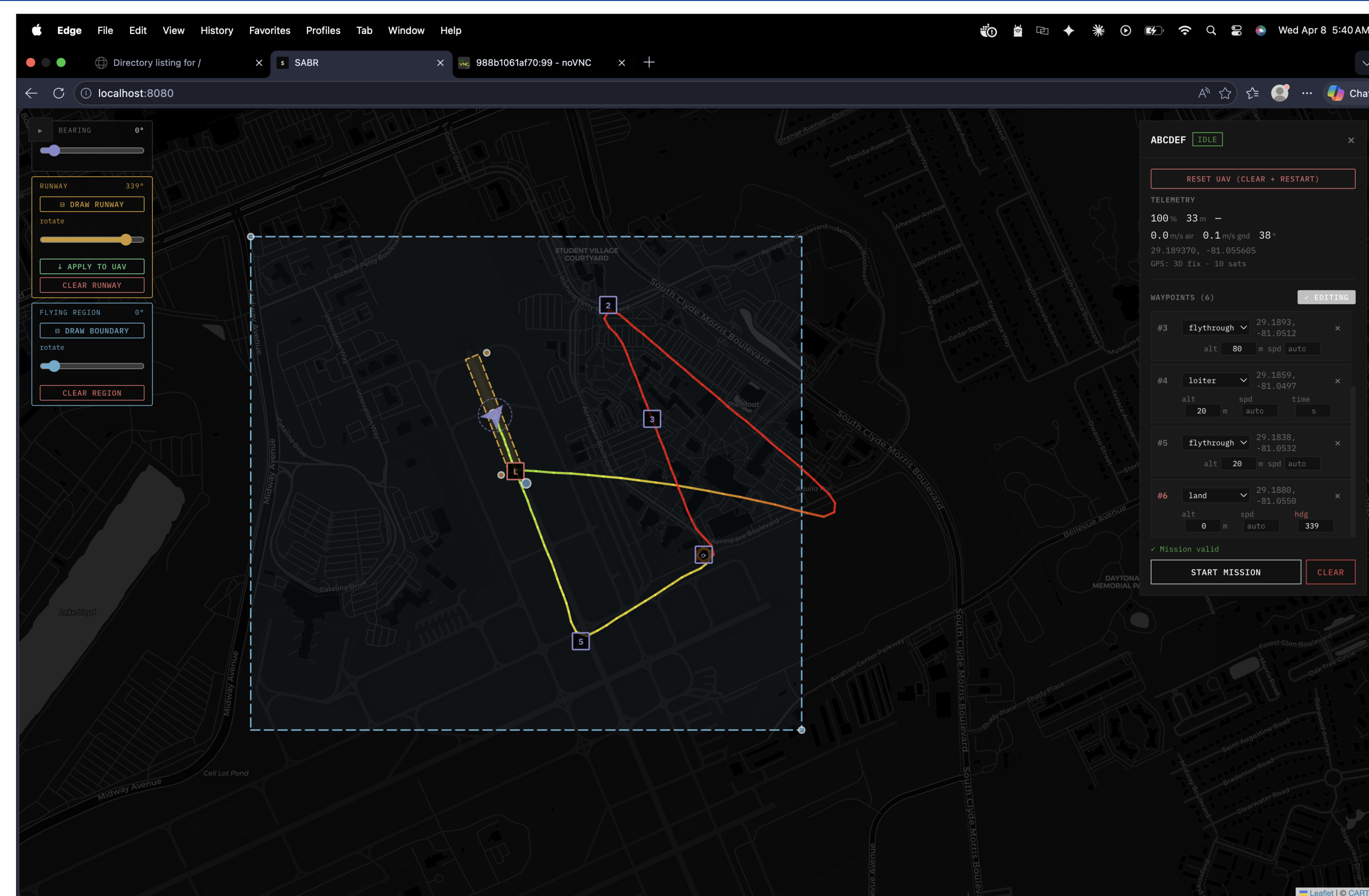


Figure 2. SABR operator UI (live screen capture).
Browser-based ground station with Leaflet map, live telemetry panel, and right-click waypoint upload.

Problem

Academic UAV stacks rely on short-range RF radios, preventing beyond-visual-line-of-sight (BVLOS) operation and limiting multi-UAV scalability.

SABR replaces RF with a Starlink-connected control plane, range is bounded only by internet reach, not radio line of sight.

Key Capabilities

- **Fly from anywhere.** An operator with a laptop and Starlink dish can plan, launch, and monitor missions from any location on the planet.
- **Fully autonomous missions.** Right-click waypoints on a map; the UAV takes off, flies the route, and lands without further input.
- **Live telemetry.** Altitude, GPS, heading, and flight phase stream to the browser in real time over the satellite link.
- **Multi-UAV ready.** One operator can command multiple aircraft from a single interface.
- **One-command deploy.** The entire ground station boots from a single terminal command, no manual setup required.

The Aircraft



Figure 3. 60 cc gas-powered fixed-wing UAV with Starlink terminal.
The SABR airframe: a 60 cc gas-turbine combustion-powered fixed-wing UAV with a Starlink dish mounted on top for satellite internet connectivity, enabling BVLOS command and live telemetry from anywhere.

How SABR Works

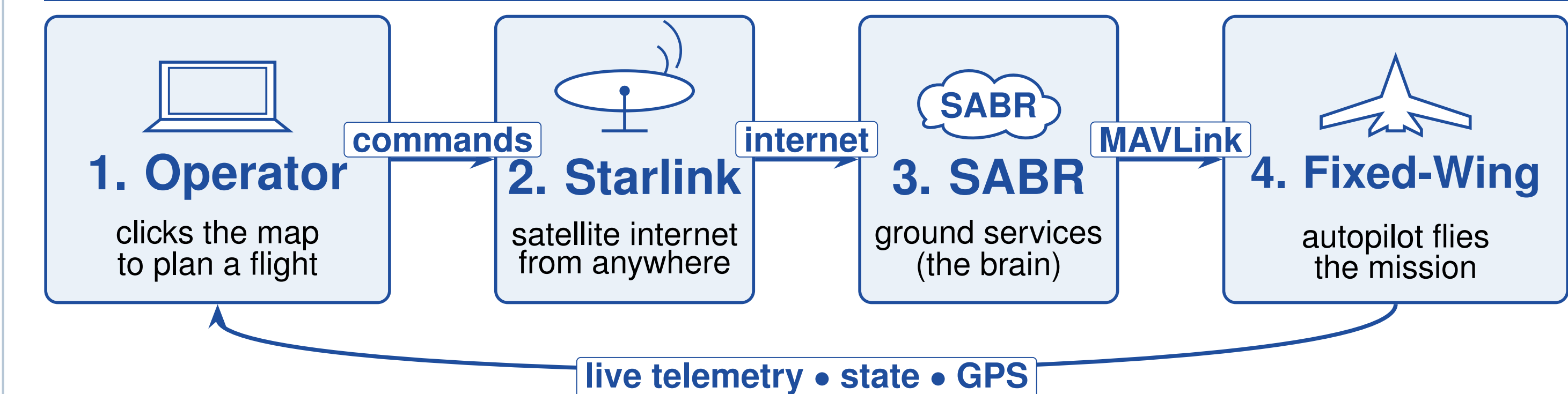


Figure 4. End-to-end SABR control loop.
Commands flow left to right; live telemetry returns along the same path.

MQTT Topic Bus

Topic	QoS	Direction
api/status	1	API → any
uav/<s>/state	1	companion → API
uav/<s>/telemetry	0	companion → API
uav/<s>/waypoints	1	API → companion

Table 1. MQTT v5 topic namespace.

All command, state, and telemetry messages flow through Mosquitto on these four topics. QoS 1 guarantees delivery for critical state and waypoint messages; QoS 0 is used for high-rate telemetry to minimize overhead. The <s> wildcard is the UAV serial number, enabling multi-aircraft operation on a single broker.

Future Work

- **Hardware build.** Move from SITL to a physical Pixhawk-class autopilot.
- **Taxi & flight tests.** Bench, taxi, then short-hop flights at an approved field.
- **Multi-UAV.** Several companions on the same bus from one operator UI.

Companion State Machine

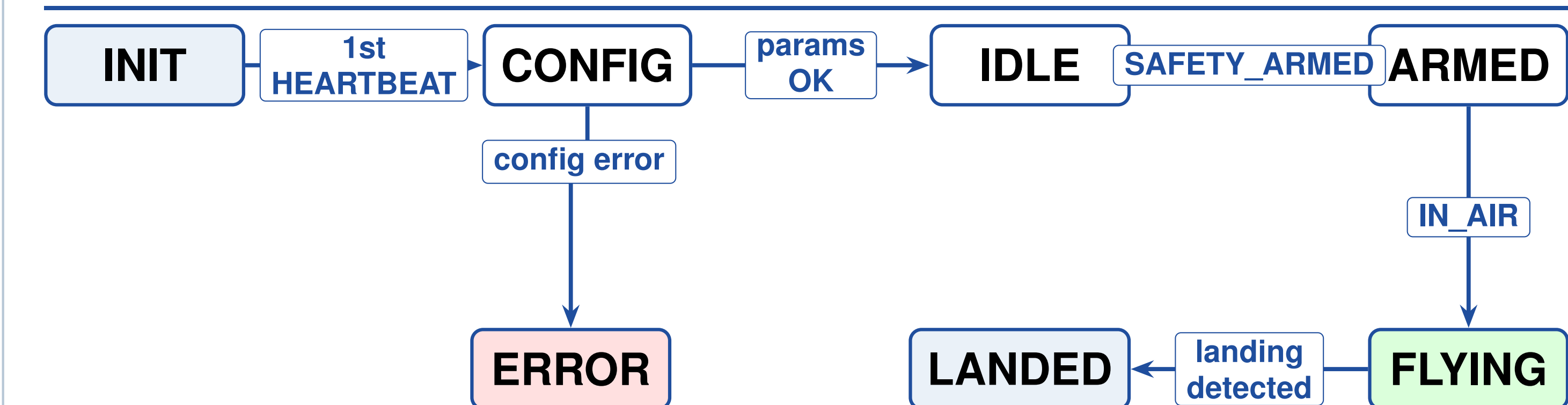


Figure 5. UAV companion state machine.
Transitions published as QoS 1 retained MQTT messages on uav/<serial>/state.